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Irradiation Induced Grain Boundary Flow—A New Creep Mechanism at the Nanoscale

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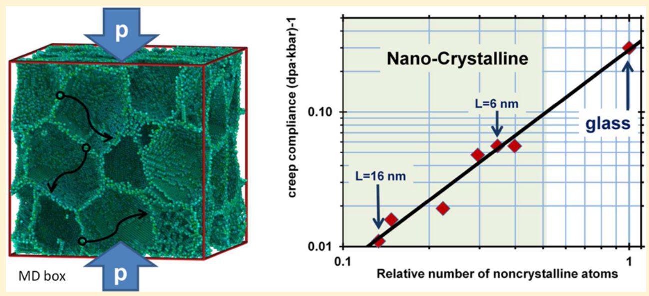
ABSTRACT: A new mechanism of irradiation enhanced creep is proposed for nanocrystalline materials. It derives from local relaxations within the grain boundaries as they absorb point defects produced by irradiation. The process is studied by inserting point defects into the grain boundaries and following the materials response by molecular dynamics. Calculated creep compliances are found in good agreement with those measured in dilute nanocrystalline Cu−W alloys [Tai, K.; Averback, R. S.; Bellon, P.; Ashkenazy Y. *Scr. Mater.* **2011**, *65*, 163]. The simulations provide a direct link between irradiation induced creep in nanocrystalline materials with radiation-induced viscous flow in amorphous materials, suggesting that grain boundaries in these materials can be treated as an amorphous phase. We provide a simple analytic model based on this assumption that reproduces the main features of the observed creep rates, a linear dependence on stress, inverse dependence of grain size, a weak dependence on temperature, and a reasonable estimate of the absolute creep rate.



KEYWORDS: *Irradiation induced creep, amorphous metals, metallic glass, atomistic simulation, nanocrystalline materials*

Irradiation of materials with energetic particles leads to the creation of large supersaturations of point defects, vacancies, and interstitials, which recombine, cluster, or flow to distributed sinks. Research over the past several decades has illustrated that these excess defects, over time, can result in a variety of detrimental effects in materials, including the degradation of mechanical properties and dimensional instabilities.[1] A straightforward strategy for suppressing the damage incurring from irradiation is simply to create a large density of sinks and traps for point defects that ultimately leads to their recombination and neutralization. Present attempts to realize such microstructures have focused on the incorporation of high densities of nanoscale features within the microstructure. The addition of nanoinclusions, such as nano-oxide dispersion-strengthened martensitic steels, is one approach.[2] Another is the synthesis of ultrafine grained materials, as these materials can now be stabilized to temperatures approaching their melting points.[3,4] Materials with small grain sizes, however, are susceptible to creep, and so a fundamental question that must be addressed in designing nanocrystalline materials for use in a reactor environment or ion-implanted nanostructures concerns their susceptibility to creep deformation under prolonged irradiation.

Creep deformation can occur by various mechanisms, depending on the temperature, applied stress, and grain size. At low stresses and small grain sizes, creep[5] derives from the net fluxes of point defects established by the variation in sink/source strengths of grain-boundaries (GBs) under either compressive or tensile tractions. Diffusion can occur either through grain boundaries[6] or the bulk[7,8] with the latter being more important in higher temperatures. At higher stresses a more effective mode of creep results from the asymmetric interaction of point defects with existing dislocations leading to preferred nucleation, glide, or climb processes.[9] In irradiated materials, this picture is modified by a supersaturation of point defects produced by the external radiation. These point defects can interact with existing dislocation structures or nucleate new ones, thereby leading to enhanced dislocation related creep rates. In nanocrystals, however, dislocations are not available for this process,[10] and unless stresses are very high, the flow of vacancies and interstitials to sinks can be considered random and thus not introducing significant creep. Moreover, since point defects are created uniformly there are no significant concentration gradients, and again no significant enhancement of diffusional creep is expected.[11,12] This all leads to the conclusion that radiation-induced creep should be highly limited in nanograined materials. Nevertheless, recent experiments show that irradiation-induced creep (IIC) does, in fact, occur in dilute nanocrystalline (NC) Cu−W alloys and that these creep rates show a linear dependence on stress.[13,14] In many respects, similar questions concerning the mechanisms of IIC in amorphous materials have been asked,[15,16] as IIC in

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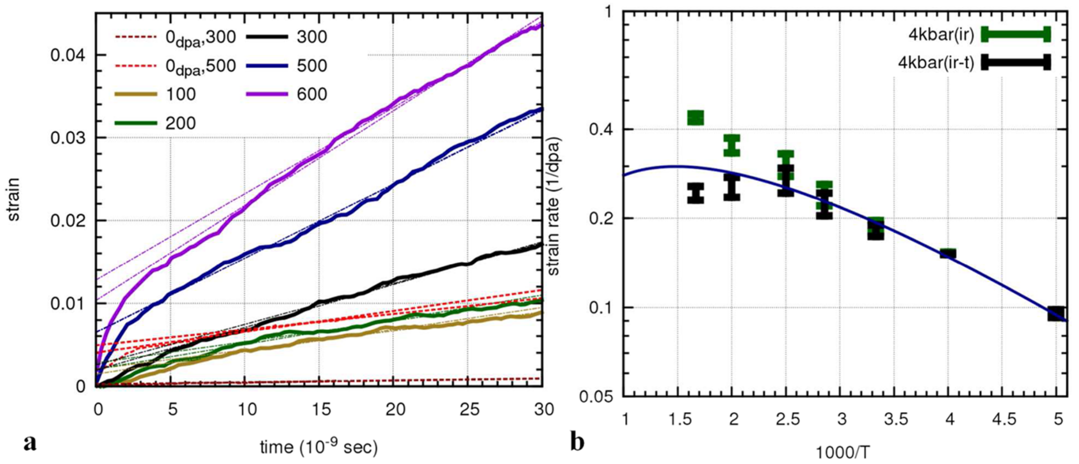

Figure 1. Temperature dependence of irradiation induced creep. (a) Strain versus time. Solid lines: with defect insertion corresponding to a dose-rate of 3×10^6 dpa/s; dashed line: without defect insertion. Maximum dose is ≈0.1 dpa. The slopes of the asymptotic lines represent steady state creep rates. (b) Strain rates (strain per dpa) in an Arrhenius plot. In green are the measured values from part a, and in black are the values corrected for thermal creep. The line is a fit to $\varepsilon' = AT^{-1}\exp(-q/kT)$ where ε' is the strain per defect with $q = 0.06 \pm 0.004$ eV and $A = 5.5 \pm 0.7$ (10^{-4}/K).

these materials is much larger than in their crystalline counterparts.[17] In this work we suggest that IIC in these two classes of materials, in fact, derives from the same basic mechanism.

Our model of IIC in nanocrystalline materials relies on the assumption that creep deformation occurs principally from the absorption of irradiation induced point defects at random locations within grain boundaries, as suggested by recent experiments.[13,14] The anisotropic flow of defects in an applied stress field is neglected, although this possibility is a simple addition to the model. This assumption greatly simplifies modeling efforts since it enables one to accelerate the dynamics of IIC by directly injecting defects into the boundaries and thus use conventional molecular dynamics to calculate the response. For subsequent comparison with experiments, we rely on standard radiation-enhanced diffusion theory[18] to estimate the rate at which point defects enter the GBs for a given irradiation flux, as discussed below. This procedure of using defect injection is similar to our past calculations of creep rates in amorphous alloys under irradiation.[19] In that case we showed that simulations employing actual recoil events gave very similar results to ones using defect injection.

NC copper samples were produced by randomly orienting copper grains on a face-centered cubic (fcc) super lattice. The centers of the grains were shifted from the superlattice points via a Gaussian function with a full-width at half-maximum (fwhm) set at 10% of the average grain radius. Atoms were then removed to create centroidal Voronoi tessellation[20] and subsequently relaxed by annealing and applying stresses up to 8 kbar in 200 ps cycles using molecular dynamics (MD) simulations. All MD calculations reported here were performed using large-scale atomic/molecular massively parallel simulator (LAMMPS),[21] with an embedded atom method (EAM) potential described in ref 22 to model pure copper. This potential reproduces both macroscopic and defect properties well (we verified, however, that results were not modified when another EAM potential for Cu was used[23]). Following sample preparation, the simulation of irradiation was carried out in cycles, where in each cycle a fixed number of defects were inserted into the grain boundaries or adjacent monolayers. The introduction of defects was achieved by either removing an atom or adding an atom. As part of this procedure we performed a short initial local relaxation through conjugate gradient energy minimization of atoms surrounding the introduced defect.[24] We note that varying the method of this initial relaxation from overdamped dynamics to conjugate gradient resulted in no significant change in the measured results.[25] Following defect insertion, the system was evolved by normal pressure and temperature (NPT) molecular dynamics.[26,27] During this dynamic stage a set of extended relaxations took place. The time required for these relaxations was on the order of a few picoseconds. We verified that outcomes were insensitive to the period of time allowed for the dynamic stage by modifying this time between 20 ps and 160 ps per cycle. Relaxation times longer than ≈20 ps assured full relaxation. Following this relaxation period we repeated the cycle by introducing a new set of defects at the grain boundaries of the evolving system. Our results therefore should be independent of dose rate for insertion rates less than $\sim 10^{10}$ (s^{-1}). We note that it is this fast relaxation that enables us to accelerate the dynamics several orders of magnitude over actual experimental displacement rates without introducing significant errors. In contrast, if IIC were simply due to enhanced Coble creep, for example, then such MD calculations would not be possible.[28]

We first report briefly on the temperature dependence of irradiation induced creep. Figure 1 shows the creep behavior for a sample with an average grain size of 6 nm, an applied stress of 4 kbar, and a defect production rate of 3×10^6 dpa/s. For comparison, the yield strength of this sample is greater than 12 kbar. The creep data show an initial primary creep stage (which can be related to initial relaxation of the sample), followed by steady state creep. We have verified that during steady state creep there is no change in the total volume of the system. The steady state creep rates, determined from Figure 1a, are shown in Figure 1b in an Arrhenius plot. Here it is observed that the temperature dependence is extremely weak, characterized by an apparent activation energy of ≈0.06 eV. The thermal creep component was calculated in a separate set of runs without defect insertion. For comparison, the creep data on the same samples and applied stress, with and without correction for thermal creep are plotted in Figure 1b. Below T = 400 K, thermal creep correction is negligible, and so to avoid the need to correct for thermal effects, we report on calculations at $T \leq$ 300 K in the remainder of this study . We verified that once

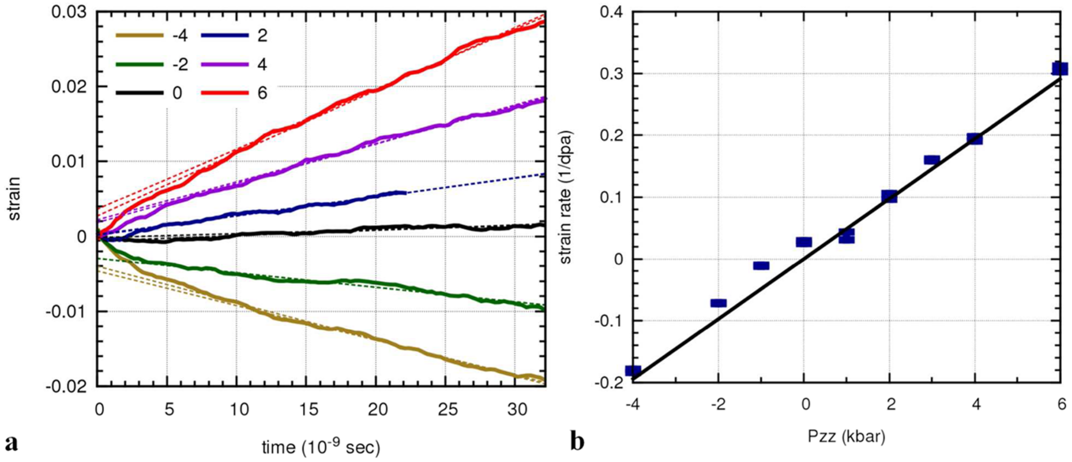

Figure 2. Pressure dependence of the measured creep. (a) Measured strain versus time for T = 300K, grain size = 6 nm, and defect production rate of 3×10^6 dpa/s. The total dose is ≈0.1 dpa. (b) Steady-state creep rates versus stress. The fit represented by $\varepsilon' = C_p\sigma_{zz}$ with $C_p = 0.048 \pm 0.002$ $(\text{dpa}\cdot\text{kbar})^{-1}$ and for a sample with an average.

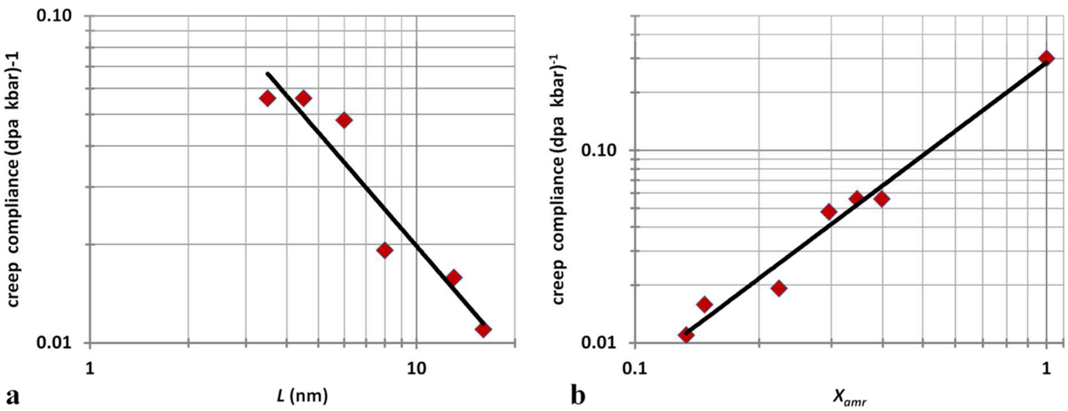

Figure 3. (a) Creep compliance at T = 300 K versus average grain size. The fit is to $\varepsilon' = CL^n$ with $n = -1.1 \pm 0.1$. (b) Same points as part a: this time the x-axis is the relative amount of atoms which are amorphous and the slope in this case $\varepsilon' = CX^n_{\text{amr}}$ with $n = 1.5 \pm 0.1$.

thermal creep is avoided the measured creep rate increases linearly with dose rate for dose rates between 6×10^6 dpa/s and 5×10^5 dpa/s; that is, the total strain depended only on the total accumulated dose.

The dependence of the irradiation induced creep rate on stress, over a wide range of stresses, is presented in Figure 2 for a sample with a NC grain size of 6 nm and a temperature of T = 300 K. Here it is seen that the creep rates are well-described by linear dependence on applied stress. The creep rates appear, moreover, symmetric around zero, with any possible threshold stress being less than ≈0.1 kbar. The creep compliance is given by the slope of the line in Figure 2b; its value is $C_p = 0.048 \pm 0.002$ $(\text{dpa}\cdot\text{kbar})^{-1}$.

The same procedure was applied for samples with other grain sizes, with the results shown in Figure 3a. These data fit well to an inverse dependence on grain size. We thus infer a generalized creep equation with explicit dependence on stress (σ), temperature (T), and grain size (L) of the form

$$\frac{d\varepsilon}{dt} = C\frac{\sigma}{kTL}\exp(-0.05/kT) \tag{1}$$

where the compliance C = 0.05 nm·K/(dpa·kbar). We note that, for samples with grain sizes of 5, 6, and 8 nm, results were unchanged by increasing the cell size by a factor of 1.5 and thus including about three times as many grains.

A comparison of eq 1 with recent experimental measurements on dilute, nanocrystalline, two-phase Cu−W alloys referred to above[13,14] shows reasonable agreement, as we now explain. First, the creep rate in these experiments showed a linearly dependence on stress, in agreement with the simulations. Second, the experimental creep rates increased with temperature between 300 K and ≈473 K but then became constant at higher temperatures. This was explained in ref 14 on the basis that, between 300 and 473 K, the defect kinetics were in the so-called recombination regime, and so many defects recombine before reaching the GBs. The fraction of defects reaching the GBs increased with increasing temperature. Above 473 K, the kinetics entered the sink-limited regime, at which point all defects that escape their nascent cascades migrate to GBs. We thus compare our results to the measurements above 473 K, since our simulations assume all defects arrive at GBs. (We can compare our results at other temperatures by correcting for recombination). In this regime, the experimental creep rate is independent of temperature, which means that both the defect kinetics is in the sink-limited regime *and* the GB creep mechanism is independent of temperature, the latter being in agreement with the simulations. The grain-size dependence was not measured in these experiments, but for the grain size of L ≈ 35 nm used in the experiments and at a temperature of 473 K, the experimental creep compliance was 3×10^{-3} (dpa·kbar)$^{-1}$. If we use eq 1 to scale the simulations to this temperature and grain size, we obtain a value of 1.2×10^{-2} (dpa·kbar)$^{-1}$ or four times of the experimental value. The simulations, of course, assume that all defects produced during irradiation arrive at grain boundaries, whereas numerous experiments and MD simulations have shown that only a small fraction ≈1−10% of the defects produced during heavy ion irradiation escape the cascade region and become *freely migrating* defects.[29,30] It has been suggested, however, that this fraction is considerably larger in nanocrystals owing to the proximity of the boundaries.[31] In light of the above description, the agreement between the experiments and simulations is thus very reasonable; it is within the uncertainty of our knowledge of the exact number of freely migrating defects, thus giving credence to the simulation model. We emphasize that, although the simulations are performed at dpa rates that are orders of magnitude greater than the experimental rates, at 473 K the system is in the sink-limited regime, and therefore the creep per unit dose (dpa) is independent of the displacement rate.[14,18] Lastly, we checked whether the nano-W particles in the experiments may have affected the experimental creep rates by performing additional simulations with comparable numbers of Nb precipitates in the GBs. No effect of the precipitates was observed (Nb was used for this test since reliable potentials are available for Cu−Nb, but not for Cu−W).

To understand the observed creep behavior more systematically, we propose a mean field model for GB creep. Before outlining the model, however, we first address the question of the mechanism of mass flow in the GBs. Some clues to answering this question derive from the magnitude of the irradiation induced creep compliance and its dependence on grain size and temperature. In Figure 3b we have replotted the grain size dependence of creep rate, but here as a function of the molar fraction of noncrystalline atoms X_{amr} (in the nanocrystalline case, this is identical to the number of grain boundary atoms).[36] Extrapolation of these data to $X_{amr} \approx 1$ yields a creep compliance of 0.3 (kbar·dpa)$^{-1}$. This value is remarkable, since it is the same value that we had obtained previously for amorphous CuTi alloys using the same defect insertion method described above. In that paper, we had pointed out that this value also agrees with the creep compliance measured experimentally on various amorphous materials, both metallic glasses and amorphous (a-)SiO_2.[19] The temperature dependence of the creep compliance in amorphous materials, moreover, was shown to be very weak, as found here for NC-Cu. To verify that the creep response found in that work for a-CuTi also represents amorphous Cu, we calculated the creep compliance for amorphous Cu, using the same methods and again obtain the values: 0.25 (dpa·kbar)$^{-1}$ at T = 10 K and 0.3 (dpa·kbar)$^{-1}$ at T = 100 K, thus showing the same magnitude and weak dependence on temperature.[37] Above T = 100 K, a-Cu crystallizes, and so higher temperatures could not be tested.

The possibility that the structure of GBs might be similar to an amorphous phase has been discussed for many years, and more recently in nc-materials using detailed MD calculations.[38,39,28] The mobility of atoms in GBs has also been shown to be quite similar to that in amorphous materials. In particular, Zhang et al. analyzed the grain boundary dynamics of Σ5 boundaries and found "stringlike" cooperative motions that resembled the dynamics observed in supercooled liquids below their glass temperatures.[32,33] They found that relative displacements have a prominent non-Gaussian component that resembled previous observations in relaxation of amorphous samples, where the non0Gaussian relaxations were shown to be spatially correlated and named "Strings".[34,35] An additional link between the relaxation in the grain boundary and behavior of amorphous materials arises from a detailed look at the relaxation events. A direct observation of the displaced atoms is presented in an inset to Figure 4, and indeed it is clear that

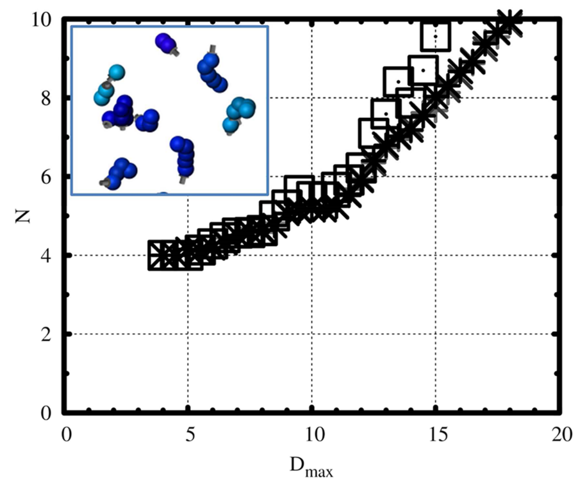

Figure 4. Average number of atoms participating in each relaxation event (N) versus the maximal size of the bounding ellipsoid (D_{max}). Data are given for relaxation events in samples with L = 6 nm, L = 8 nm, L = 16 nm, and amorphous as dash, plus, x, and rectangles accordingly. As an inset a single set of relaxation events is shown for a sample with L = 8 nm. Only atoms that moved more than half the nearest neighbor distance are plotted (this was verified to be larger than the edge of the Gaussian thermal displacement edge). The color shade indicates the distance from the crystal front, and arrows connect initial to final positions.

most of these displaced atoms are arranged in one-dimensional clusters. We identified groups of atoms that moved together over the relaxation period following defect introduction and fitted this cluster with an ellipsoid. A plot of the number of displaced atoms versus the semimajor axis of the ellipsoid, given in Figure 4, gives another indication that indeed these relaxation events are resulting in one-dimensional "string" like events. We further note that the resulting is plot is practically identical for all samples with 6, 8, and 16 nm grain size as well as for a fully amorphous Cu sample. We defer additional analysis of these displacements to a later time.

Our macroscopic model for IIC in nanocrystalline materials is built on the assumptions that the grain boundaries consist of an amorphous phase and that the introduction of a point defect into the boundary results in a localized change in the shape of this material due to the applied stress, σ. For simplicity we consider cubical grains of side L. We assume that the thickness of the grain boundary is a result of local minima in energy and therefore GB relaxations associated with the injection of point defects do not lead to expansion or shrinkage of the GB phase but rather to reshaping of the crystalline grains. This is consistent with refs 5 and 39 and was further verified in our simulations as the number of amorphous-like GB atoms does not change from the defect injection process. A series of these local relaxation events can lead to creep as there is an energetic preference for relaxation in accordance with the external applied stress. The number of atomic volumes crossing a plane normal to the direction of the flow per relaxation event is given by

$$\frac{\Delta N}{\Delta n_r} = A_r \lambda \nu \frac{1}{2}[\sigma(z + \lambda) - \sigma(z)]/\Omega \tag{2}$$

where ΔN is the number of atomic volumes added as a consequence of (Δn_r) relaxation events; ν is the effective strain modulus, A_r is the cross sectional area of the transformation zone, λ is the length of the transformation zone, z is a coordinate along the GB, $\sigma(z)$ is the stress in the GB plane at coordinate z, and Ω is the atomic volume.

Assuming λ is small, we write

$$\frac{\Delta N}{\Delta n_r} = \frac{A_r \lambda^2 \nu}{2\Omega}\left[\frac{\sigma(z + \lambda) - \sigma(z)}{\lambda}\right] = \frac{A_r \lambda^2 \nu}{2\Omega}\frac{\partial \sigma}{\partial z} \tag{3}$$

The flux of atoms through the grain boundary at a coordinate z as a result of these relaxation events is given by $J \propto (1/A_{gb}(z))(\partial n_r/\partial t)(\partial N/\partial N_r)$ where $A_{gb}(z)$ is the grain boundary area at a coordinate z, and $(\partial n_r/\partial t)$ is proportional to the irradiation flux which we assume to be constant. We therefore obtain $J(z) \propto (1/A_{gb}(z))(\partial \sigma/\partial z)$, and due to continuity we have that $(\partial J/\partial t) = (\partial^2 J/\partial z^2)$ (assuming there are no further sources and sinks within the grain boundary, i.e., the grains maintain their shape). At steady state the local current does not change, and since $A_{gb}(z)$ is assumed constant, $(\partial^2\sigma/\partial z^2)$ = Const and so $(\partial \sigma/\partial z) \propto z$. Assuming that the gradient of the stress is zero at the center of the grain boundary, that stress is zero at the corners, and that the average of the stress in the grains is equal to the applied (constant) stress, we obtain $\sigma(z) = 6\sigma_0 z/L(1 - z/L)$ and $\langle(\partial\sigma/\partial z)\rangle = K(\sigma_0/(L/2))$. This is identical to the expression found previously for material flowing through grain boundaries[13,40] where K is a geometrical factor equal to 3/2 for cubes of side L. We thus have from eq 3, $(\Delta N/\Delta n_r) = K(A_r \lambda^2 \nu \sigma_0/\Omega L)$.

The total increase in the sample length after an irradiation creating ϕ relaxation events per atom is

$$\Delta L = \frac{\Delta N}{\Delta n_r}\cdot\left(\phi \frac{L^{3\cdot}}{A_r \lambda}\right)\cdot\left(\frac{\Omega}{L^2}\right) = K\Omega\lambda\nu\sigma_0\phi$$

where ϕ is the ion dose in displacements per atom (dpa), $\phi L^3/(A_r\lambda)$ is the total number of relaxation events, and Ω/L^2 is the length change for one atom being added to the grain boundary.

Thus, $(d\varepsilon/d\phi) = (1/\phi)(\Delta L/L) = (K\lambda\nu\sigma_0/L)$ and with the approximation $K \approx 1$, this leads to

$$\frac{d\varepsilon}{d\phi} = \frac{\lambda\nu\sigma_0}{L} \tag{4}$$

The creep rate, with respect to dose, is thus linear in stress, inversely proportional to grain size and essentially independent of temperature.

For a bulk amorphous material, each strain increment gained from a relaxation event is simply additive,[15,16] and we can immediately write $(d\Delta/d\phi) = \nu\sigma_0$. The ratio between the radiation induced fluidity of the amorphous material and that of a polycrystalline with an average grain size of L is then $K\lambda/L \approx \lambda/L$. If we compare the MD determined compliance for the amorphous material 0.3 $(\text{dpa}\cdot\text{kbar})^{-1}$ with the value of 0.05 $(\text{dpa}\cdot\text{kbar})^{-1}$ for NC Cu with L = 6 nm, we obtain λ = 1 nm. This value is consistent with the length scale of the "string" like nature of the relaxation volume illustrated shown in Figure 4. Furthermore, a direct examination of the displacement histogram as a result of the relaxation events shows a strong correlation between the non-Gaussian displacements in the amorphous case and those of atoms within the grain boundaries.

In summary we have shown using MD simulations that nanocrystalline metals are susceptible to irradiation induced grain boundary creep and that this creep is due to localized relaxations in the grain boundaries. Quantitative agreement was obtained between the simulations and recent measurements of irradiation-induced creep in Cu−W alloys. The simulation further showed that, as the grain size becomes small, the creep rates extrapolate to those found in amorphous materials. A simple analytical model was developed that rationalizes these simulations, and it provides a direct link establishing how creep in an amorphous-like GB phase controls the ICC response in NC metals.

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Notes

The authors declare no competing financial interest.

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(25) Relaxation through overdamping is done for 50 fs at a rate which is equivalent to reducing atoms velocity by 10% each femtosecond, so that in all our simulations the kinetic energy at the end of this stage is equivalent to a thermal energy which is less than $T = 2$ K. We verified that results were not modified if the local relaxation was done by locally modifying atom coordinates via conjugate gradient minimization. At the end of this stage the whole system is equilibrated at fixed volume to the original temperature using NVT thermostat within 1 ps.
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